

NAME:

NO:

CLASS:

RIVERSIDE SECONDARY SCHOOL



PRELIMINARY EXAMINATION 2026

SUBJECT : MATHEMATICS
SUBJECT CODE /PAPER : 4052/02
LEVEL/STREAM : Sec 4 Express & 5 Normal (Academic)
DURATION : 2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

If working is needed for any question, it must be shown with the answer.

Omission of essential working will result in loss of marks.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to 1 decimal place.

For π , use either your calculator value or 3.142, unless the question requires the answer in terms of π .

At the end of the examination fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 90.

Submit Part A and Part B separately.

Mathematical Formulae*Compound interest*

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$\text{Standard deviation} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2}$$

Part AAnswer **all** the questions.

1 (a) Simplify $\frac{19a^6b^0c^2}{35c^5} \div \frac{9a^2b^4}{14c}$.

Answer [2]

(b) Simplify $\left(\frac{12x^{-4}y^6}{27}\right)^{-\frac{1}{2}}$ and write your answer in positive index notation.

Answer [2]

(c) Given that $2^{x+1} = 2^x + 32$, find x .

Answer $x =$ [2]

- (d) Write as a single fraction in its simplest form $\frac{x+2}{x-3} - \frac{60}{2x^2-18}$.

Answer [4]

- (e) Explain why, for all integer values of n , $n(n^2-1)$ is always even.

Answer

.....

..... [2]

- 2 (a) A quadratic equation has solutions $x = p$ and $x = -2p$, where p is a positive integer. Write the equation in the form $x^2 + bx + c = 0$, where b and c are expressed in terms of p .

Answer [2]

- (b) A quadratic curve intersects the x -axis at $x = 2 + \sqrt{a}$ and $2 - \sqrt{a}$.
Find the equation of its line of symmetry.

Answer [1]

- (c) (i) A straight line passes through $P(-5, 9)$ and $Q(-9, 3)$.
Find the column vector $\overrightarrow{PQ}$.

Answer $\overrightarrow{PQ} = \dots\dots\dots$ [2]

- (ii) Write down another column vector that is parallel to $\overrightarrow{PQ}$.

Answer [1]

- 3 (a) During the World Cup, merchandise such as jerseys and caps are sold.

Game 1 sold 12 000 jerseys and 4000 caps.

Game 2 sold 9000 jerseys and 5600 caps.

Game 3 sold 11800 jerseys and 3000 caps.

The number sold, in thousands, are shown in the matrix

$$\mathbf{W} = \begin{pmatrix} 12 & 4 \\ 9 & 5.6 \\ 11.8 & 3 \end{pmatrix}$$

A jersey is priced at \$80 and a cap at \$30.

- (i) Represent the price of the merchandise in a 2×1 column matrix $\mathbf{P}$.

Answer $\mathbf{P} =$ [1]

- (ii) Evaluate the matrix $\mathbf{T} = \mathbf{WP}$.

Answer $\mathbf{T} =$ [2]

- (iii) Explain what the elements in $\mathbf{T}$ represent.

Answer
 [1]

- (b) The remaining stock consists of 7000 jerseys and 8000 caps.

After a storewide discount was applied, the sale of the remaining stock is represented by the matrix equation

$$d \begin{pmatrix} 7 & 8 \end{pmatrix} \begin{pmatrix} 80 & 0 \\ 0 & 30 \end{pmatrix} = \begin{pmatrix} 448 & 192 \end{pmatrix},$$

where d is a constant.

Find the percentage discount given by the store.

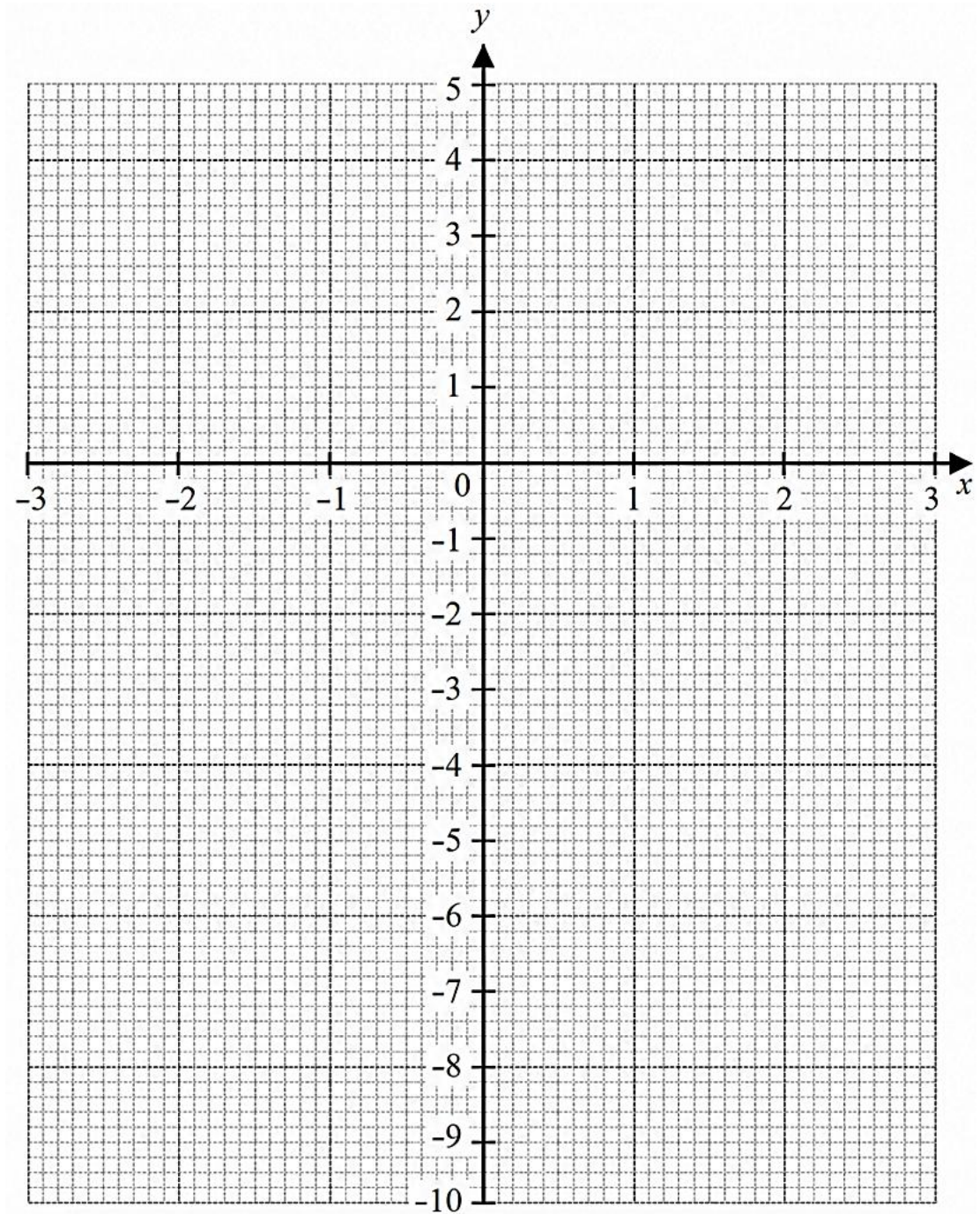
Answer% [4]

- 4 (a) (i) Complete the table of values below for $y = \frac{1}{2}x^3 - \frac{1}{2}x^2 - 2x + 2$.

x	-3	-2	-1	0	1	2	3
y	-10	0		2	0	0	5

[1]

- (ii) On the grid, draw the graph of $y = \frac{1}{2}x^3 - \frac{1}{2}x^2 - 2x + 2$ for $-3 \leq x \leq 3$.



[3]

- (iii) The water quality level of a pond over a 3-week period is modelled by the curve $y = \frac{1}{2}x^3 - \frac{1}{2}x^2 - 2x + 2$, where x represents the time in weeks.

The minimum safe level for fish to survive is given by the line $y = 2 - x$.

On the same grid, draw the line $y = 2 - x$.

[1]

- (iv) From the graph, write down the value(s) of x where the water quality equals the minimum safe level.

Answer $x = \dots\dots\dots$ [2]

- (v) The fish are safe when the water quality is above the minimum safe level. Use your graph to find the range of values of x for which the fish are safe.

Answer $\dots\dots\dots$ [1]

- (b) The graph of $y = ka^x$, where k and a are positive integers, passes through the points $(0, 4)$ and $(3, 32)$. Find the value of k and of a .

Answer $k = \dots\dots\dots$ [2]

$a = \dots\dots\dots$ [1]

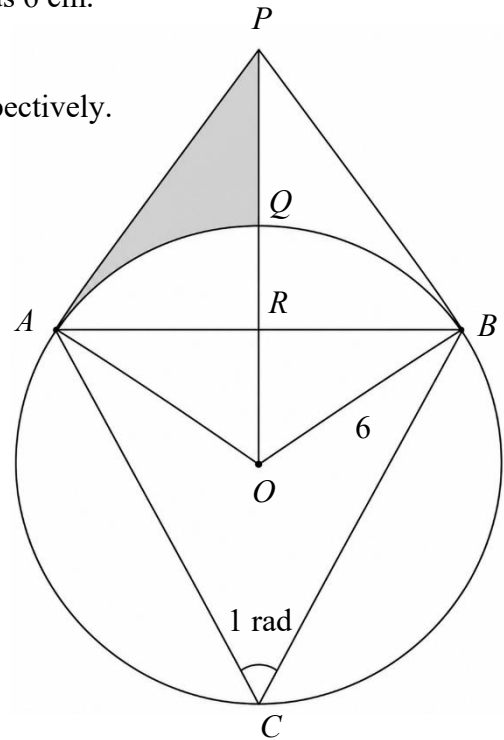
- 5 The diagram shows a circle with centre O and radius 6 cm.

A, Q, B and C lie on the circle.

PA and PB are tangents to the circle at A and B respectively.

O, R, Q and P lie on a straight line.

Angle $ACB = 1$ radian.



- (a) Show that triangle AOP is congruent to triangle BOP .
Give a reason for each statement you make.

Answer

.....

.....

.....

.....[3]

- (b) State the value of angle AOB in radians.
Give a reason for your answer.

Answer Angle $AOB = \dots\dots\dots$ radians

Reason:[2]

- (c) Find the area of the shaded region.

Answer cm^2 [4]

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Part B continues on Page 13

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RIVERSIDE SECONDARY SCHOOL**PRELIMINARY EXAMINATION 2026****SUBJECT : MATHEMATICS****SUBJECT CODE /PAPER : 4052/02****LEVEL/STREAM : Sec 4 Express & 5 Normal (Academic)****Part B**Answer **all** the questions.

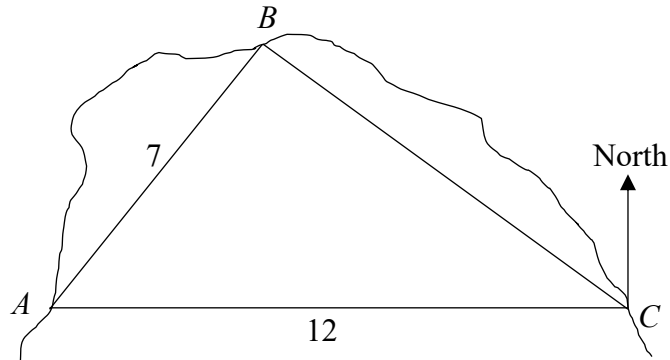
Part BAnswer **all** the questions.

- 6 Triangle ABC is the boundary of a nature reserve.
 A , B and C are three points on level ground.

$$AB = 7 \text{ km.}$$

C is 12 km due east of A .

The area of the nature reserve is 33 km^2 .



- (a) (i) Show that angle $BAC = 51.8^\circ$.

Answer

[3]

- (ii) Calculate the length of BC .

Answerkm [2]

- (iii) the bearing of B from C .

Answer° [3]

- (b) An observation deck is 60 m above point B .

- (i) A path is to be built from B to a point on AC , across the nature reserve.
Find the shortest possible length of the path.

Answerkm [2]

- (ii) A visitor is at the top of the observation tower.
Find the greatest angle of depression from the top of the tower to a point on AC .

Answer [2]

- 7 Ryan cycles 4 km from his home to a convenience store at an average speed of x km/h. He spends 20 minutes in the store before cycling back home along the same route, at an average speed of 2.5 km/h slower. The total time taken, including the time spent in the store, is 1 hour.

(a) Show that $2x^2 - 29x + 30 = 0$.

Answer

[5]

- (b) Solve the equation $2x^2 - 29x + 30 = 0$.
Give your solutions correct to three decimal places.

Answer $x = \dots\dots\dots$ or $\dots\dots\dots$ [3]

- (c) Explain why one of your answers in **part (b)** cannot be accepted.

Answer $\dots\dots\dots$
 $\dots\dots\dots$ [1]

- (d) Calculate Ryan's average speed when cycling home from the convenience store.

Answer $\dots\dots\dots$ km/h [2]

- 8 In a Hyrox Fitness Challenge, the official finish times of 150 participants from Regent Polytechnic were recorded.

(a) The table shows the time, t minutes, taken by the participants.

Time, t minutes	$50 < t \leq 70$	$70 < t \leq 90$	$90 < t \leq 110$	$110 < t \leq 130$	$130 < t \leq 150$
Frequency	12	29	x	39	21

(i) Find the value of x .

Answer $x = \dots\dots\dots$ [1]

(ii) Calculate an estimate of the mean time taken.

Answer $\dots\dots\dots$ mins [1]

(iii) Calculate an estimate of standard deviation of the times taken.

Answer $\dots\dots\dots$ mins [1]

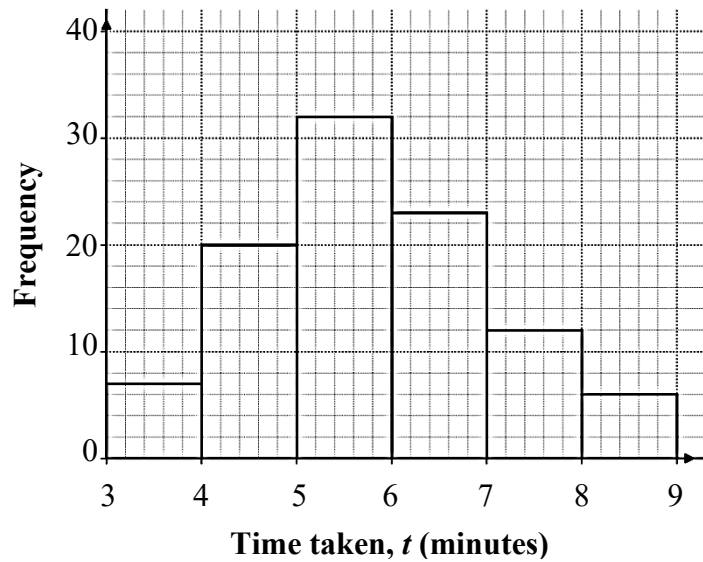
(iv) Another set of 150 participants from Summit Polytechnic have an estimated mean finish time of 99.5 minutes and standard deviation of 25.6 minutes.

A commentator says that participants from Summit Polytechnic performed better than those in Regent Polytechnic.

Explain why this comment is not fully justified.

Answer $\dots\dots\dots$
 $\dots\dots\dots$
 $\dots\dots\dots$
 $\dots\dots\dots$ [2]

- (b) The histogram shows the distribution of time, t minutes, taken by 100 participants in a SkiErg challenge.



A participant who completes the SkiErg Challenge in at most 5 minutes is called a fast finisher.

- (i) Two participants are selected at random for an interview.
Find the probability that both participants are fast finisher.

Answer [2]

- (ii) Three participants are selected randomly for a photoshoot.
Find the probability that at least one of them is a fast finisher.

Answer [2]

- (iii) A coach claims that the number of participants who completed the SkiErg Challenge in less than 5.5 minutes is greater than the number who took more than 6.5 minutes.

Explain why his conclusion cannot be made from the histogram.

Answer

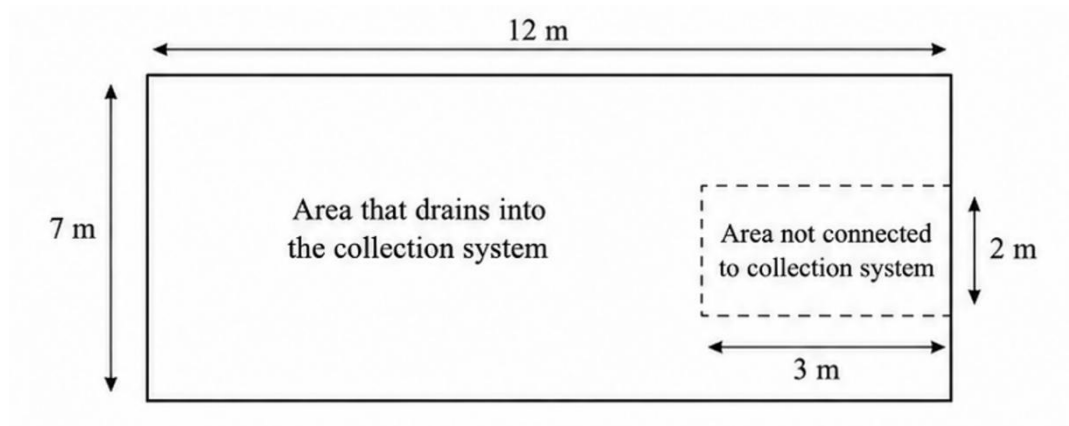
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.....[1]

- 9 A team at EcoFlow Dynamics is designing a rainwater collection system to water its vegetable garden.

The diagram shows the roof that drains into the rainwater collection system.

The rectangular section shown by the dashed lines is not connected to the collection system.



Top-down view of roof

It is known that 1 mm of rain falling on 1 m^2 of roof produces 1 litre of water.

During a heavy shower, 15 mm of rain falls evenly on the roof.
Only 82% of the rainwater falling on the collecting area is collected.

The vegetable garden requires 170 litres of water each day.
The tank must provide enough water for 5 consecutive days without rain.
For safety reasons, the amount of water in the tank must not exceed 90% of its capacity.

The team has a total budget of \$800, which includes the cost of the tank and its installation.

Use $\pi = 3.142$.

Tank	Shape and internal dimensions	Cost of tank	Installation cost
A	Cylinder of diameter 1.0 m and height 1.1 m	\$620	\$130
B	Cuboid of length 1.1 m, width 1.0 m and height 1.1 m	\$680	\$120
C	Cylinder of diameter 1.2 m and height 0.9 m	\$710	\$80

- (a) Calculate the amount of water collected during the heavy shower.

Answerlitres [4]

- (b) Using the information given and your answer to **part (a)**, suggest a suitable tank for the rainwater collection system.

Justify your decision with calculations.

Answer

[7]